

Quickest Change Detection via Cloud MIMO Radar with Time Delay-Quantization

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Abstract—This paper introduces a novel method for quickly detecting a change in respiration rate (RR) from a normal to an unknown abnormal state using quickest change detection (QCD) with cloud multiple-input multiple-output (MIMO) radar. In contrast to quantizing radar returns, we propose to estimate and quantize local time-varying delays caused by respiration prior to transmission to fusion center (FC), thereby reducing communication load while retaining physiological information. However, at the FC, unknown post-change RR, time-varying delays, and quantization introduce high-dimensional and discrete characteristics, reshaping the statistical structure of MIMO radar data, limiting the effectiveness of conventional QCD methods. We propose a quantized delay based generalized likelihood ratio test (QDGLRT) at FC for effective data transmission and abnormal RR detection. A closed-form asymptotic statistic is derived to facilitate performance analysis. Numerical simulations validate the theoretical analysis and demonstrate the effectiveness of the proposed method for real-time abnormal RR detection.

Index Terms—Abnormal respiratory rate monitoring, generalized likelihood ratio test (GLRT), performance analysis, quantized quickest change detection (QuQCD).

I. INTRODUCTION

Abnormal respiratory rate (RR) is a critical indicator of various health conditions, and its timely detection is essential for effective diagnosis and intervention [1]. Non-contact sensing methods, particularly radar-based monitoring, have recently emerged as promising solutions for unobtrusive respiratory assessment [2], [3]. Among them, cloud multiple-input multiple-output (MIMO) radar systems exploit spatial diversity from distributed local sensors connected to a fusion center (FC) via a wireless backhaul network, where limited local processing reduces transmission burden and enables efficient cooperative detection [4]–[6]. Such architectures enable enhanced abnormal RR detection sensitivity but also require efficient communication and signal processing techniques to handle large data volumes [7]. To facilitate efficient wireless communication with the FC [8], local sensors typically quantize received measurements before transmission [9].

Quickest change detection (QCD) provides a statistical framework for identifying abrupt changes in observation streams with minimal delay and has been widely studied in signal processing and anomaly detection [10]–[13]. Nevertheless, its application to distributed radar-based physiological

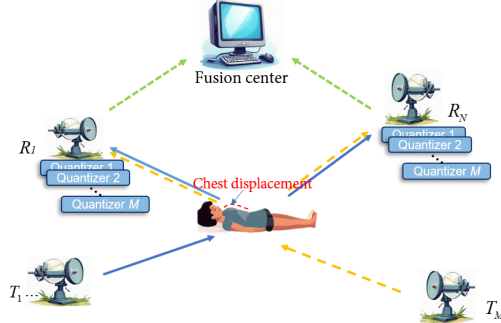


Fig. 1. Schematic diagram of QuQCD for abnormal respiratory rate monitoring in a cloud MIMO radar system.

monitoring remains underexplored [14]. In particular, detecting abnormal RR with quantized cloud MIMO radar presents several new challenges [7]. Since quantization not only reduces communication overhead but also fundamentally reshapes the statistical structure of the data, raising open questions about achievable detection performance under resource constraints [6]. Time-varying respiration induces high-dimensional and nonlinear distortions in radar returns, and the post-change abnormal RR is typically unknown, posing significant challenges to the direct application of conventional QCD frameworks in radar sensing environments.

By analyzing the time-varying delays in cloud MIMO radar echoes induced by respiration, rather than directly quantizing the received signals at the local sensors as in our previous work [7], the RR-related delays are first estimated and quantized locally, and then sequentially transmitted to the FC, where a quantized delay based generalized likelihood ratio test (QDGLRT) is proposed to detect the abnormal RR change. The test statistic of QDGLRT is derived from discrete quantization outputs and often lacks closed-form expressions. To address this, a Gaussian approximation is employed to characterize the quantization output, thereby yielding a closed-form solution for the QDGLRT test statistic and providing valuable insights into performance analysis.

II. SIGNAL MODEL AND TIME-DELAY QUANTIZATION

We consider a cloud MIMO radar composed of M transmitters and N receivers, spatially distributed for monitoring the respiratory rate f_r of a subject. The m -th transmitter and the n -th receiver are located at $(x_{t,m}, y_{t,m})$ and $(x_{r,n}, y_{r,n})$, respectively, where $m = 1, \dots, M$ and $n = 1, \dots, N$. The detection task focuses on identifying an unknown change in f_r , specifically changes from a normal rate $f_r = f_{nr} \in \mathfrak{R}_n$

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to an abnormal rate $f_r = f_{ar} \notin \mathfrak{R}_n$. Here, $\mathfrak{R}_n$ represents the admissible range of normal respiration. At time kT_s ($k = 1, \dots, K$), the lowpass equivalent time-sampled version of the signal emitted by m -th ($m = 1, \dots, M$) transmitter is $E/Ms_m(kT_s)$, where E denotes the total energy and T_s is the sampling interval. The transmitted waveforms are mutually orthogonal and preserve orthogonality under delay [15]. The signal received at the n -th ($n = 1, \dots, N$) local receiver from the m -th transmitter, after reflection from the target chest, is

$$\begin{aligned} r_{mn}[k] &= \sqrt{E/M} \xi_{mn} s_m(kT_s - \tau_{mn}[k, f_r]) + w_{mn}[k] \\ &= s_{mn}[k, f_r] + w_{mn}[k], \end{aligned} \quad (1)$$

where the ξ_{mn} is the complex reflection coefficient, and $w_{mn}[k]$ is the clutter-plus-noise on the mn -th path, assumed to be zero-mean complex Gaussian with variance σ_u^2 . The time delay $\tau_{mn}[k, f_r]$ associated with the mn -th path is affected by the subject's respiration and can be expressed as [2]

$$\tau_{mn}[k, f_r] = \frac{d_{mn} + \mu_{mn}[k, f_r]}{c}, \quad (2)$$

where d_{mn} is the nominal distance, and $\mu_{mn}[k, f_r]$ represents the periodic chest displacement caused by respiration, which is modeled as [16]

$$\mu_{mn}[k, f_r] = 2a_{r_{mn}} \sin(2\pi f_r k T_s), \quad (3)$$

with $a_{r_{mn}}$ and f_r denote the respiratory amplitude and frequency, respectively.

A. Local Time Delay Estimation and Quantization

In a cloud MIMO radar system, local sensors transmit their data to the FC through a backhaul network, where the respiration change detection is conducted. To reduce the communication burden, the local data are quantized prior to transmission.

Note that $\tau_{mn}[k, f_r]$ in (2) is a function of the unknown RR and encapsulates all respiration-related information. Moreover, it can be independently obtained at each local sensor. To reduce the communication load, quantizing this scalar is more efficient than directly quantizing the complex received signals $r_{mn}[k]$. Therefore, it is reasonable to estimate and quantize $\tau_{mn}[k, f_r]$ at the local sensor prior to transmission to the FC.

Thus, as per (1), the likelihood functions of $r_{mn}[k]$ is

$$p(r_{mn}[k]; f_r) = \frac{1}{\pi \sigma_u^2} \exp \left\{ -\frac{1}{\sigma_u^2} |r_{mn}[k] - s_{mn}[k, f_r]|^2 \right\}.$$

The maximum likelihood (ML) estimate of the $\tau_{mn}[k, f_r]$ is obtained by solving the following maximization problem

$$\hat{\tau}_{mn}[k, f_r] = \arg \left\{ \max_{\tau_{mn}[k, f_r]} \ln p(r_{mn}[k]; f_r) \right\}. \quad (4)$$

Then the $\hat{\tau}_{mn}[k, f_r]$ at local sensor is quantized as

$$q_{mn}[k] = \mathbb{Q} \{ \hat{\tau}_{mn}[k, f_r] \} = \begin{cases} U_0, \gamma_0 < \hat{\tau}_{mn}[k, f_r] < \gamma_1 \\ U_1, \gamma_1 < \hat{\tau}_{mn}[k, f_r] < \gamma_2 \\ \vdots \\ U_{D-1}, \gamma_{D-1} < \hat{\tau}_{mn}[k, f_r] < \gamma_D, \end{cases}$$

where $D = 2^b$ is quantization levels, with b representing the number of quantization bits. Each level corresponds to a value $U_d \in \{U_0, \dots, U_{D-1}\}$, determined by the range of $\hat{\tau}_{mn}[k, f_r]$ within the quantizer thresholds $\gamma_0, \gamma_1, \dots, \gamma_D$. For a uniform quantizer with dynamic range $[\tau_{\min}, \tau_{\max}]$, the thresholds are

specified as $\gamma_0 = \tau_{\min}, \gamma_D = \tau_{\max}$ with step size $\Delta = (\tau_{\max} - \tau_{\min})/2^b$.

We focus on the problem of detecting an abnormal RR change, where the RR f_r undergoes a change from a normal state $f_r = f_{nr}$ to an abnormal state $f_r = f_{ar}$ at an unknown change time k^* . The normal RR f_{nr} can typically be determined from historical measurements [17], whereas the abnormal RR f_{ar} is unknown due to the absence of prior knowledge. After local quantization, the abnormal RR change detection problem at FC can be formulated as

$$H_0 : \quad \mathbf{q}_r[k] = [q_{11}[k], q_{12}[k], \dots, q_{MN}[k]]^\dagger, \quad (5)$$

$$q_{mn}[k] = \mathbb{Q} \{ \hat{\tau}_{mn}[k, f_{nr}] \} \quad \forall k,$$

$$H_1 : \quad \mathbf{q}_r[k] = [q_{11}[k], q_{12}[k], \dots, q_{MN}[k]]^\dagger, \quad (6)$$

$$q_{mn}[k] = \begin{cases} \mathbb{Q} \{ \hat{\tau}_{mn}[k, f_{nr}] \}, & k < k^*, \\ \mathbb{Q} \{ \hat{\tau}_{mn}[k, f_{ar}] \}, & k \geq k^*, \end{cases}$$

where the $k^* \geq 1$ is the unknown change time.

III. SUDDEN ABNORMAL RR DETECTION AT FUSION CENTER

To promptly identify the change in the RR, based on the received quantized time-delay measurements and unknown post-change f_{ar} , the FC employs a sequential detection scheme, i.e., QDGLRT based QuQCD. Focusing on the analysis of quantization effects, the backhaul between local sensors and the fusion center is assumed ideal [4], [5], neglecting latency, fading, and other impacts.

A. QDGLRT based QuQCD Method

The estimate $\hat{\tau}_{mn}[k, f_{nr}]$ and $\hat{\tau}_{mn}[k, f_{ar}]$ in (5) and (6) are ML estimators of $\tau_{mn}[k, f_{nr}]$ and $\tau_{mn}[k, f_{ar}]$, respectively. Under standard regularity conditions and in the moderate-to-high SNR regime [18], they are asymptotically Gaussian with means $\tau_{mn}[k, f_{nr}]$ and $\tau_{mn}[k, f_{ar}]$ and variance

$$\begin{aligned} \sigma_{mn, f_r}^2 &= -\mathbb{E} \left\{ \nabla_{\tau_{mn}[k, f_r]} [\nabla_{\tau_{mn}[k, f_r]} \ln p(r_{mn}[k]; f_r)] \right\} \\ &= 1 / \left(\frac{2}{\sigma_u^2} \left| \frac{\partial s_{mn}[k, f_r]}{\partial \tau_{mn}[k, f_r]} \right|^2 \right), \end{aligned} \quad (7)$$

where $f_r = f_{nr}$ or f_{ar} . Therefore, the probabilities of $q_{mn}[k]$ in (5) taking the quantization levels $U_{d_{mnk}}$ (for $d_{mnk} = 1, \dots, D-1$) under H_0 can be given by

$$\begin{aligned} P_0(q_{mn}[k]) &= P(q_{mn}[k] = U_{d_{mnk}}; f_{nr}) \\ &= \left[Q \left(\frac{\gamma_{d_{mnk}} - \tau_{mn}[k, f_{nr}]}{\sigma_{mn, f_{nr}}} \right) - Q \left(\frac{\gamma_{d_{mnk}+1} - \tau_{mn}[k, f_{nr}]}{\sigma_{mn, f_{nr}}} \right) \right], \end{aligned} \quad (8)$$

and under H_1 after the change can be given by

$$\begin{aligned} P_1(q_{mn}[k]) &= P(q_{mn}[k] = U_{d_{mnk}}; f_{ar}) \\ &= \left[Q \left(\frac{\gamma_{d_{mnk}} - \tau_{mn}[k, f_{ar}]}{\sigma_{mn, f_{ar}}} \right) - Q \left(\frac{\gamma_{d_{mnk}+1} - \tau_{mn}[k, f_{ar}]}{\sigma_{mn, f_{ar}}} \right) \right], \end{aligned} \quad (9)$$

where $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty \exp(-t^2/2) dt$ represents the tail probability of standard normal distribution.

Then, at FC, the likelihood functions of $\mathbf{q}_r[k]$ in (5) and (6) at time k are

$$H_0 : \quad P_0(\mathbf{q}_r[k]) = \prod_{n=1}^N \prod_{m=1}^M P_0(q_{mn}[k]) \quad \forall k, \quad (10)$$

$$H_1 : \begin{cases} P_0(\mathbf{q}_r[k]) = \prod_{n=1}^N \prod_{m=1}^M P_0(q_{mn}[k]), k < k^*, \\ P_1(\mathbf{q}_r[k]) = \prod_{n=1}^N \prod_{m=1}^M P_1(q_{mn}[k]), k \geq k^*. \end{cases} \quad (11)$$

For abrupt RR change detection, based on (10) and (11), the LLR at the FC at time $n_c \geq k^*$, parameterized by the unknown k^* and f_{ar} , is given by

$$\begin{aligned} \mathcal{G}_{k^*, f_{ar}}^{n_c} &= \ln \frac{\prod_{k=1}^{k^*-1} P_0(\mathbf{q}_r[k]) \prod_{k=k^*}^{n_c} P_1(\mathbf{q}_r[k])}{\prod_{k=1}^{n_c} P_0(\mathbf{q}_r[k])} \\ &= \ln \frac{\prod_{k=1}^{k^*-1} \prod_{n=1}^N \prod_{m=1}^M P_0(q_{mn}[k]) \prod_{k=k^*}^{n_c} \prod_{n=1}^N \prod_{m=1}^M P_1(q_{mn}[k])}{\prod_{k=1}^{n_c} \prod_{n=1}^N \prod_{m=1}^M P_0(q_{mn}[k])} \\ &= \sum_{k=k^*}^{n_c} \sum_{n=1}^N \sum_{m=1}^M \ln \frac{P_1(q_{mn}[k])}{P_0(q_{mn}[k])} = \sum_{k=k^*}^{n_c} L_{k, f_{ar}}. \end{aligned} \quad (12)$$

In this context, the LLR is high-dimensional and discrete, and its statistical properties, determined by the quantized time-delay measurements in (8) and (9), are time-varying and depend on the unknown post-change RR. This dynamic behavior, induced by the unknown change in the RR, goes beyond the conventional QCD framework. Consequently, the QDGLRT is proposed to address the complex and time-varying abnormal RR detection problem.

At the FC, the test statistic of the QDGLRT for abnormal RR change detection is defined as

$$\mathcal{T}_{E, n_c} = \max_{1 \leq k^* \leq n_c} \mathcal{G}_{k^*, \hat{f}_{ar, k^*}}^{n_c} = \max_{1 \leq k^* \leq n_c} \sum_{k=k^*}^{n_c} L_{k, \hat{f}_{ar, k^*}}, \quad (13)$$

where the ML estimate of the abnormal RR is

$$\hat{f}_{ar, k^*} = \arg \left\{ \sup_{f_{ar} \notin \mathfrak{R}_n} \mathcal{G}_{k^*, f_{ar}}^{n_c} \right\}, \quad (14)$$

where "sup" is the supremum operator, and the estimation is obtained by combining a grid search with the Nelder-Mead algorithm [18]. The system alarm an abnormal RR when the test statistic $\mathcal{T}_{E, n_c}$ exceeds the threshold h_E for the first time. Then, the abnormal RR alerting time is defined as

$$\hat{n}_E = \inf \{n_c \mid \mathcal{T}_{E, n_c} \geq h_E\}, \quad (15)$$

where the h_E is determined by a specified false alarm level.

B. Performance Analysis

The goal is to promptly identify the abnormal respiration change. Once an abnormal alarm is declared at time n_s by (15), the detection performance is evaluated under two hypotheses. Under the null hypothesis H_0 (keep normal RR), the mean time to false alarm (MTFA)

$$\mathcal{F}_0 = \mathbb{E}_0\{n_s\}, \quad (16)$$

measures the expected duration before a false alarm occurs. In contrast, under the hypothesis H_1 (abnormal RR present), the detection is evaluated by the average detection delay (ADD) $\mathcal{A}_1 = \mathbb{E}_1\{(n_s - k^*)^+\}$ [19]. To provide a robust performance

guarantee against worst-case scenarios, the worst-case ADD (WADD) is adopted as a key metric,

$$\mathcal{W}_1 = \sup_{k^* \geq 1} \text{ess sup} \mathbb{E}_1\{(n_s - k^*)^+ \mid \{\mathbf{q}_r[1], \dots, \mathbf{q}_r[k^*]\}\}, \quad (17)$$

where $\mathbb{E}_j\{\cdot\}$ ($j \in \{0, 1\}$) denotes the expectation under H_0 and H_1 , and "essup" is the essential supremum. With a pre-specified constraint on false alarms, denoted by Γ , the abnormal respiration detection problem can be formulated as the following optimization problem

$$\min \mathcal{W}_1 \quad (18)$$

$$\text{subject to } \mathcal{F}_0 \geq \Gamma, \quad (19)$$

where h_E is determined under H_0 to satisfy the $\mathcal{F}_0 \geq \Gamma$

For abnormal RR detection, quantizing the estimated time delays results in discrete observations and a high-dimensional QuQCD statistic whose distribution lacks a closed-form expression, making direct performance analysis intractable.

To simplify the analysis, a uniform quantizer with step size Δ is employed. An approximate description for the quantization is adopted to provide analytical insight into how quantization parameters affect the statistics of the quantized signal. Furthermore, to provide the best achievable performance bound, we consider a special case in which the f_{ar} is assumed to be known. The following lemma provides the quantization approximation.

Lemma 1 [6]: *When a symmetric quantizer with an uniform quantization step Δ is applied to a Gaussian input x with mean μ and variance σ^2 , the output $q = x + \epsilon$, where ϵ represents the quantization error. The probability density function (pdf) of q is given by*

$$p(q) = \frac{e^{-\frac{1}{2(\sigma^2 + \Delta^2/12)}(q-\mu)^2}}{\sqrt{2\pi(\sigma^2 + \Delta^2/12)}}, \text{ as } \Delta \rightarrow 0, \quad (20)$$

implying that q follows $\mathcal{N}(\mu, \sigma^2 + \Delta^2/12)$.

Applying the Lemma 1, the distribution $p_0(q_{mn}[k])$ of quantized delay $q_{mn}[k]$ in (5) is

$$q_{mn}[k] \sim \mathcal{N}(\tau_{mn}[k, f_{nr}], \sigma_{mn, f_{nr}}^2 + \Delta^2/12) \quad (21)$$

and the distribution $p_1(q_{mn}[k])$ of quantized observation $q_{mn}[k]$ in (6) after k^* is

$$q_{mn}[k] \sim \mathcal{N}(\tau_{mn}[k, f_{ar}], \sigma_{mn, f_{ar}}^2 + \Delta^2/12). \quad (22)$$

From (21) and (22), the corresponding LLR can then be written as

$$\begin{aligned} \tilde{L}_{k^*}^{n_c} &= \ln \frac{\prod_{k=1}^{k^*-1} \prod_{n=1}^N \prod_{m=1}^M p_0(q_{mn}[k]) \prod_{k=k^*}^{n_c} \prod_{n=1}^N \prod_{m=1}^M p_1(q_{mn}[k])}{\prod_{k=1}^{n_c} \prod_{n=1}^N \prod_{m=1}^M p_0(q_{mn}[k])} \\ &= \sum_{k=k^*}^{n_c} \tilde{C}_{k, asy}, \end{aligned} \quad (23)$$

where

$$\begin{aligned} \tilde{C}_{k, asy} &= \sum_{n=1}^N \sum_{m=1}^M \left[\frac{1}{2} \ln \left(\frac{\sigma_{mn, f_{nr}}^2 + \Delta^2/12}{\sigma_{mn, f_{ar}}^2 + \Delta^2/12} \right) \right. \\ &\quad \left. + \frac{(q_{mn}[k] - \tau_{mn}[k, f_{nr}])^2}{2(\sigma_{mn, f_{nr}}^2 + \Delta^2/12)} - \frac{(q_{mn}[k] - \tau_{mn}[k, f_{ar}])^2}{2(\sigma_{mn, f_{ar}}^2 + \Delta^2/12)} \right]. \end{aligned} \quad (24)$$

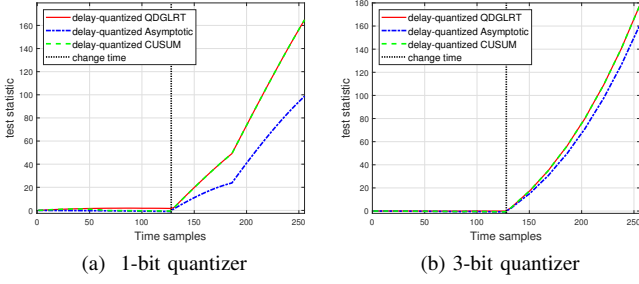


Fig. 2. A realization of test statistics for QDGLRT, asymptotic QDGLRT, and CUSUM method with known f_{ar} under different quantization levels, using a 2×4 MIMO radar. The results are shown for 256 samples with a change taking place at the 129th sample.

From (23), the asymptotic QDGLRT test statistic can then be expressed as a recursive form

$$\mathcal{T}_{Easy, n_c} = \max_{1 \leq k^* \leq n_c} \tilde{L}_{k^*}^{n_c} = \max_{1 \leq k^* \leq n_c} \sum_{k=k^*}^{n_c} \tilde{C}_{k,asy} \quad (25)$$

$$= \max\{0, \mathcal{T}_{Easy, n_c-1}\} + \tilde{C}_{n_c,asy}. \quad (26)$$

The asymptotic alerting time is defined as

$$\tilde{n}_{Easy} = \inf \left\{ n_c < k^* \mid \mathcal{T}_{Easy, n_c} \geq \tilde{h}_E \right\}, \quad (27)$$

where $\tilde{h}_E$ is determined by false alarm constraint.

With the closed-form statistic in (27), the impact of quantization parameters on the test statistic is explicit, and performance bounds can follow directly from classical CUSUM results [20], which is infeasible for (15).

IV. NUMERICAL RESULTS

This section presents numerical simulations to validate the proposed QDGLRT method. The cloud MIMO radar system has $M = 2$ transmitters and $N = 4$ receivers, placed 2 meters from a subject at (0.5, 0.5) m in a two-dimensional plane to monitor a sudden change in the RR. Each chirp uses a frequency-modulated continuous wave (FMCW) signal $s_m(kT_s) = e^{j2\pi((f_0 + m\Delta f)kT_s + \frac{1}{2}\beta(kT_s)^2)}$, with the start frequency $f_0 = 77$ GHz, frequency slope $\beta = 64.019$ MHz/ μ s, sample rate $f_s = 5.12 \times 10^6$ Hz, pulse width $T' = 67 \times 10^{-6}$ s, and inter-transmitter offset $\Delta f = 2/T'$. The analog-to-digital converter collects 256 samples per chirp. The normal RR $f_{nr} = 0.3$ Hz $\in \mathfrak{R}_n$ changes to an abnormal RR $f_{ar} = 1$ Hz $\notin \mathfrak{R}_n$ at the 129th sample. The normal RR range $\mathfrak{R}_n = [0.2, 0.35]$ Hz is defined based on age, gender, and historical data [21]. The f_{ar} is initialized via a grid search over [0.1, 1.5] Hz with a 0.05 Hz step, and Nelder-Mead refines it with a 10^{-6} convergence tolerance. The dynamic range of the uniform quantizer is $\gamma_0 = \min_{m,n} (\min_k (\tau_{mn}[k])) - \sigma_{mn}^2 = 1.3333 \times 10^{-8}$ and $\gamma_D = \max_{m,n} (\max_k (\tau_{mn}[k])) = 1.3335 \times 10^{-8}$ [22]. The clutter-plus-noise ratio (SCNR) is defined as $\text{SCNR} = 10 \log_{10}(\sum_{m=1}^M E_m / \sigma_u^2)$, with $E_m = 1/M$ and $\sigma_u^2 = 0.1$. Results are averaged over 2000 Monte Carlo simulations.

In Fig. 2, we plot realizations of the test statistics for the QDGLRT, its asymptotic version, and the optimal CUSUM method with known f_{ar} under 1-bit and 3-bit quantization. In the delay-quantized scenario, the QDGLRT exhibits performance nearly identical to that of the CUSUM in both (a) and

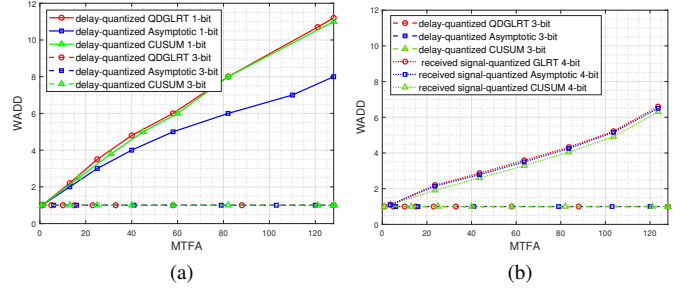


Fig. 3. WADD versus MTFA for the QDGLRT, its asymptotic approximation, and the CUSUM method with known f_{ar} using a 2×4 MIMO radar: (a) with 1-bit and 3-bit quantizers; (b) with received signal quantization [7] and time delay quantization.

(b). The proposed asymptotic approximation aligns well with the QDGLRT at 3 bits. The statistic values increase slightly from 1 to 3 bits, with little further improvement thereafter, suggesting that 3-bit quantization is sufficient.

In Fig. 3(a), we evaluate the WADD in (18) versus the MTFA in (19) for the QDGLRT, its asymptotic version, and the optimal CUSUM method with known f_{ar} under 1-bit and 3-bit quantization. As the quantization level increases, all methods exhibit enhanced detection performance. At 3-bit quantization, each method demonstrates the capability to promptly detect abnormal respiratory events. The QDGLRT achieves detection performance comparable to that of the optimal CUSUM method under 1-bit and 3-bit quantization, demonstrating the effectiveness of the proposed method. Furthermore, under 3-bit quantization, the asymptotic WADD closely aligns with that of the QDGLRT, since the reduced quantization step size provides a more accurate approximation. Though Lemma 1 assumes $\Delta \rightarrow 0$, it accurately predicts performance trends even with 3-bit quantization. As shown in Fig. 3(b), the WADDs obtained by 4-bit received signal quantization [7] are compared with that achieved using 3-bit delay quantization. Despite using a higher quantization resolution for the received signal, received signal quantization results in a larger WADD than delay quantization. This result arises from the significantly smaller dynamic range required by the proposed delay-quantization method, which further confirms its effectiveness in detecting weak signals such as abnormal respiration.

V. CONCLUSION

Abnormal RR detection using cloud MIMO radar was investigated for the first time based on a QuQCD method. By analyzing the time-varying delays in cloud MIMO radar echoes caused by respiration, we estimate and quantize the delays locally before transmitting them to the FC, and propose a QDGLRT detector for efficient data transmission and abnormal RR detection. By applying a Gaussian approximation to characterize the quantized delays, closed-form expressions for the test statistic were derived to facilitate performance analysis. Numerical studies validated the effectiveness of the proposed QDGLRT method, demonstrating the superior performance over signal-quantization based QCD and highlighting its potential for non-contact abnormal respiration monitoring.

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